

Crownpath (Epic Seven) Scout Engine - Improvement Report

Repo: <https://github.com/elsBlue/CP> | Analyzed via GitHub tarball (no git clone) | HEAD main @ 6cd73615 | Generated 2026-09-06 (WIB)

Ringkasan singkat (Bahasa Indonesia)

Crownpath adalah aplikasi scout arena Epic Seven. Data hero (roles/tags/kit/unique) di `src/lib/e7/heroes.ts` masuk ke engine di `engine.ts`: `classifyDefense` mengklasifikasi tembok lawan ke salah satu dari 8 archetypes, `wallThreats` membangun daftar ancaman, lalu `recommendCounters` mengisi resep dari `recipes.ts` menjadi lineup counter.

Katalog: 213 hero; 100 verified (46.9%), 113 belum verified. Scout mengabaikan kit unverified saat scoring. Ada 8 tipe dinding, 8 seed recipes, 6 preset defense, ~111 threat keys.

Verdict Wall=8: YA - batas keras di tipe/data (`ArchetypeId` union + `ARCHETYPE_IDS` + skor + `ARCHETYPE_META`), sekaligus keputusan produk tertulis: `VERIFY_HERO.md` menyebut "Eight wall types is the ceiling" dan `AGENTS.project.md` "Wall types stay at eight". Bukan bottleneck slice(0,8) di runtime; bukan sekadar komentar chat. Perluasan taxonomy sengaja dibekukan selama fokus verify hero.

Prioritas: (1) verifikasi unit prefer/SAMPLE unverified, (2) normalisasi effects (91 unverified kosong), (3) kurangi hardcoded `idSet.has` di `classifyDefense`, (4) bersihkan prefer lists, (5) multi-label / archetype baru hanya setelah coverage verified naik.

1. Architecture: data -> engine -> scout output

`heroes.ts` + `recipes.ts` + `threats.ts`

- > `catalog.ts` (Zustand; `normalizeHero` via `effects.ts`)
- > optional DB overlay `api.ts` `ensureCatalog/loadCatalog`
- > `scout-view.tsx` (arena 4 / GW 3 + roster pool)
- > `classifyDefense(enemyIds)`: verified-only scores for 8 `ArchetypeIds` + `wallThreats`
- > `recommendCounters`: `recipesFor` + `fillRecipe` + score; up to 4 unique teams
- > UI: Type + Watch + Lineups (why/setup/pitfalls/gaps)

Key file map

Path	Role
<code>src/lib/e7/heroes.ts</code>	Hero catalog (213). verified/checkedAt/kit/uniques. Floor 3000 lines.
<code>src/lib/e7/types.ts</code>	<code>ArchetypeId</code> (exactly 8), Role, Tag, Recipe, DefenseRead.
<code>src/lib/e7/recipes.ts</code>	<code>ARCHETYPE_META</code> , 8 RECIPES, 6 PRESET_DEFENSES.
<code>src/lib/e7/threats.ts</code>	<code>wallThreats</code> / <code>unansweredThreats</code> / <code>lineupLimitNote</code> .
<code>src/lib/e7/engine.ts</code>	<code>classifyDefense</code> + <code>recommendCounters</code> + <code>jobFor/whyFor/pitfallsFor</code> .
<code>src/lib/e7/effects.ts</code>	<code>deriveEffects</code> / <code>normalizeHero</code> when <code>effects[]</code> empty.
<code>src/lib/e7/catalog.ts</code>	Client catalog store; <code>getHero/allHeroes/allRecipes</code> .
<code>src/lib/e7/api.ts</code>	Server seed/sync heroes; admin CRUD; match/wall stats.
<code>src/lib/e7/store.ts</code>	Arena/GW enemy boards, roster, <code>restrictToRoster</code> .
<code>src/components/e7/scout-view.tsx</code>	Scout UI wiring <code>classifyDefense</code> + <code>recommendCounters</code> .
<code>src/lib/e7/VERIFY_HERO.md</code>	Verify protocol; forbids adding <code>ArchetypeId</code> (ceiling=8).
<code>src/lib/e7/SCOUT_AUDIT.md</code>	Copy-grounding checklist for Why/Setup/Breaks-if.
<code>AGENTS.project.md</code>	Core-file floors; Wall types stay at eight.
<code>scripts/guard-e7-core.mjs</code>	Guard against truncated core matcher files.
<code>migrations/0002+*.sql</code>	<code>heroes/recipes/presets/matches</code> ; verified/checked_at/rarity/speed.

2. Findings

Heroes coverage

- Total: 213; verified 100 (46.9%); unverified 113

- Verified tiers: SS 10, S 49, A 31, B 10
- Unverified tiers: S 21, A 54, B 38
- Elements: fire 52, earth 45, ice 44, light 36, dark 36
- Classes: warrior 51, mage 42, ranger 36, soulweaver 29, knight 28, thief 27
- checkedAt: 2026-08-31 (50), 2026-09-01 (31), 2026-09-06 (19)
- All heroes have numeric baseSpeed
- Kit avg: verified ~298 chars; unverified ~258
- Verified without uniqueEffects: diene
- Empty effects[]: verified 4, unverified 91 (fallback deriveEffects)
- UNSURE kits: ~45 heroes
- soulburn tag on 212/213 (low signal)
- Brace mix: 82? 2-space, 131? 4-space openings

Verified special roles

- speedcap: harsetti only; soulblock: belian only; evasion: setsuka, remnant-violet
- revive: ruele-of-light, school-nurse-yulha, genesis-ras, apocalypse-ravi, maid-chloe, spirit-eye-celine, lisette
- All 100 verified heroes have jobFor cases (0 missing)

Walls

Exactly 8 Archetypelds: speed-cleave, harsetti-stall, revive-wall, injury-grind, evasion-counter, turn2-control, immunity-soul, bruiser-mix. classifyDefense scores only these; bruiser-mix baseline score=1 (catch-all). 8 recipes + 6 presets. recommendCounters caps output teams at 4 (not walls).

Threats

- ~111 distinct wallThreats keys
- unansweredThreats + lineupLimitNote for fatal gaps (Speed cap, First cycle, Revive/reset, Notos)
- Unverified wall heroes ignored for watches except Unverified kit note

Inconsistencies

- SAMPLE unverified: krau, milim, flan, peira, landy, roana, senya
- STARTER unverified: peira, iseria, flan, milim, krau, senya, landy, choux, seaside-bellona, tywin, destina
- prefer[] unverified (16): basar, destina, ilynav, iseria, jenua, krau, landy, lethe, lidica, lilibet, peira, ravi, roana, senya, shalltear, tamarinne
- Hardcoded unverified ids in engine/threats: eye-of-the-abyss-fumyr, ivana, kayron, lilibet
- RGB bases often unverified while ML forms verified
- SCOUT_AUDIT union false-positive risk still structural

3. Wall=8 verdict + evidence

VERDICT: YES - hard code/type limit + explicit product ceiling (expansion freeze)

Not a soft runtime slice. Not merely prior chat. Expanding requires Archetypeld union, ARCHETYPE_IDS, ARCHETYPE_META, classifyDefense scores/priority, recipe.vs, and VERIFY_HERO policy.

Evidence paths

1. src/lib/e7/types.ts - Archetypeld union + ARCHETYPE_IDS length 8 (~52-71)
2. src/lib/e7/recipes.ts - ARCHETYPE_META 8 keys; RECIPES length 8
3. src/lib/e7/engine.ts classifyDefense - 8-key scores; bruiser-mix baseline 1
4. src/lib/e7/VERIFY_HERO.md:16 - "Eight wall types is the ceiling"
5. AGENTS.project.md:30 - "Wall types stay at eight"
6. Contrast: recommendCounters uniqueTeams.length >= 4 break caps OUTPUT lineups, not walls

Prior Grok "8 is enough for now" matches written ceiling during hero-verify workflow: temporary product freeze encoded as hard schema limit - not accidental bottleneck.

4. Prioritized improvements

DATA P1 (quick): Verify prefer/SAMPLE/STARTER gaps

Why: Engine ignores unverified; prefer names 16 unverified ids. Impact: Better slot fills. Approach: VERIFY_HERO.md on prefer_unverified then SAMPLE/STARTER leftovers.

DATA P2 (quick): UNSURE kits + empty effects

Why: ~45 UNSURE kits; 91 empty effects[]. Impact: Cleaner watches/answerEffects. Approach: Journal pass; fill effects/buffs/debuffs.

DATA P3: Normalize rarity + formatting

Why: Mixed rarity omission; brace indentation mix. Impact: Safer DB sync/parsers. Approach: Codemod; always set rarity via heroRarity.

ENGINE P1 (quick): Prefer-list hygiene gate

Why: prefer can point at unverified (-20 slotScore; pool drop). Impact: Immediate lineup quality. Approach: Assert prefer verified in check:e7.

ENGINE P2: Soften idSet.has taxonomy

Why: Hardcoded ids miss new verified heroes. Impact: Scalable classification. Approach: roles/tags/unique detectors; id allowlists for exceptions only.

ENGINE P3 (larger): Multi-signal wall read, keep 8 labels

Why: Single winner collapses hybrids; bruiser-mix sink. Impact: Better recipe injection without new Archetypeld. Approach: Secondary scores / watch-driven boosts; revisit 9th type after verified>70%.

ENGINE P4 (larger): Grounding tests

Why: Ungrounded copy is SCOUT_AUDIT bug class. Impact: Fewer false revive/nullifier lines. Approach: Fixture walls asserting unique removal removes sentence.

5. Quick wins vs larger refactors

Quick wins (days)

- Verify or remove 16 prefer-unverified heroes
- check:e7 assert prefer/SAMPLE/STARTER ids exist (+ verified preferred)
- Fill effects[]; clear UNSURE when confirmed
- Document wall=8 ceiling for agents
- Normalize heroes.ts braces

Larger (weeks)

- Tag/role-driven classification replacing most idSet.has tables
- Secondary recipe injection for hybrids (Sanctuary, Death Sentence)
- Automated SCOUT_AUDIT fixtures in CI
- Only then consider 9th Archetypeld (types+meta+scores+recipes+policy)
- Lean on api admin upsert; keep heroes.ts as seed source of truth

6. Top 5 improvements

1. DATA: Verify/gate prefer + SAMPLE/STARTER unverified heroes

2. DATA: Clear UNSURE kits; backfill effects[]
3. ENGINE: Prefer-list hygiene + verified-first assertions
4. ENGINE: Replace brittle id hardcodes with tag/unique detectors
5. ENGINE: Multi-signal recipe selection under 8-archetype ceiling; defer new ArchetypeId

7. Method

Used `curl code.load.github.com/elsBlue/CP/tar.gz/refs/heads/main` (REST API rate-limited; gh unauthenticated). No git clone.